

DAY 4: PARTIAL DERIVATIVES AND GRADIENT - PROBLEMS

PROBLEMS IN NEWTONIAN MECHANICS

Instructions

Solve all problems showing complete work. All questions are based on Newtonian mechanics and require partial derivatives and gradient vector concepts as covered in the notes.

Problem 1: Potential Energy of a Block on a Slope

A block of mass $m = 3$ kg rests on a frictionless inclined plane. The height of the block above the ground is given by $h(x, y) = H - ax - by$, where $H = 10$ m, $a = 0.2$, $b = 0.1$, and x, y are horizontal coordinates in meters. The gravitational potential energy is $U = mgh$.

- Find $\frac{\partial U}{\partial x}$ and $\frac{\partial U}{\partial y}$.
- Find the force components $F_x = -\frac{\partial U}{\partial x}$ and $F_y = -\frac{\partial U}{\partial y}$.
- What is the magnitude and direction of the force on the block?
- In what direction would the block accelerate if released from rest at $(2, 3)$?
- Find the gradient vector ∇U at $(2, 3)$. What does its direction tell you?

Problem 2: Gravitational Potential Energy Near Earth's Surface

The gravitational potential energy of a mass $m = 5$ kg near Earth's surface is:

$$U(x, y, z) = mgz$$

where $g = 9.8$ m/s², z is height above ground in meters, and x, y are horizontal coordinates.

- Find $\frac{\partial U}{\partial x}$, $\frac{\partial U}{\partial y}$, and $\frac{\partial U}{\partial z}$.
- Interpret each partial derivative physically. Why are two of them zero?
- Find the force vector $\mathbf{F} = -\nabla U$. Show that it points straight down with magnitude mg .
- A ball is thrown upward with initial vertical velocity 10 m/s from $(0, 0, 0)$. Using the chain rule, find $\frac{dU}{dt}$ at the instant of launch.
- What does the sign of $\frac{dU}{dt}$ tell you about whether the ball is gaining or losing potential energy?

Problem 3: Particle in a 2D Harmonic Oscillator

A particle of mass $m = 0.2$ kg moves in a 2D potential:

$$U(x, y) = \frac{1}{2}k_1x^2 + \frac{1}{2}k_2y^2$$

with $k_1 = 8$ N/m and $k_2 = 18$ N/m.

- Find the force components $F_x = -\frac{\partial U}{\partial x}$ and $F_y = -\frac{\partial U}{\partial y}$.
- Find the gradient vector ∇U at the point $(0.2, 0.1)$ m.
- In what direction from this point does the potential energy increase most rapidly? What is that rate of increase?
- If the particle is released from rest at $(0.2, 0.1)$, in what direction will it initially accelerate? Explain using the relationship $\mathbf{F} = -\nabla U$.
- Write Newton's second law equations: $m\ddot{x} = F_x$, $m\ddot{y} = F_y$. What type of motion do these equations predict?

Problem 4: Marble on a Curved Surface

A marble rolls on a surface whose height (in meters) is given by:

$$h(x, y) = 5 - 0.1x^2 - 0.2y^2$$

The gravitational potential energy of the marble (mass $m = 0.1$ kg) is $U = mgh$.

- Find $\frac{\partial h}{\partial x}$ and $\frac{\partial h}{\partial y}$.
- Find the gradient of the height ∇h at the point $(1, 2)$. In what direction is the surface steepest?
- The force on the marble is $\mathbf{F} = -mg\nabla h$. Find the force components at $(1, 2)$.
- If the marble starts from rest at $(1, 2)$, in what direction will it begin to roll? Explain.
- Find the point where $\nabla h = 0$. What does this tell you about the surface at that point?

Problem 5: Central Force Potential

A planet of mass $m = 6 \times 10^{24}$ kg orbits a star. The gravitational potential energy is:

$$U(r) = -\frac{GMm}{r}$$

where $r = \sqrt{x^2 + y^2 + z^2}$, $G = 6.67 \times 10^{-11}$ N·m²/kg², and $M = 2 \times 10^{30}$ kg.

- Find $\frac{\partial U}{\partial x}$ in terms of x , y , z , and r . (Remember $r = \sqrt{x^2 + y^2 + z^2}$)
- Show that $\frac{\partial U}{\partial x} = \frac{GMmx}{r^3}$.
- Find the force components $F_x = -\frac{\partial U}{\partial x}$, $F_y = -\frac{\partial U}{\partial y}$, $F_z = -\frac{\partial U}{\partial z}$.
- Show that the force vector $\mathbf{F}$ points radially inward and has magnitude GMm/r^2 .
- At the point $(3 \times 10^{11}, 4 \times 10^{11}, 0)$ m, find the force vector. What is its magnitude?

Problem 6: Potential Energy with a Saddle Point

A particle moves in a potential energy surface given by:

$$U(x, y) = x^2 - y^2 \quad (\text{in Joules})$$

- a. Find the force components $F_x = -\frac{\partial U}{\partial x}$ and $F_y = -\frac{\partial U}{\partial y}$.
- b. Find the gradient vector ∇U at the point $(1, 1)$.
- c. In what direction from $(1, 1)$ does the potential energy increase most rapidly? Decrease most rapidly?
- d. If a particle is placed at $(1, 1)$ and given a tiny push in the $+x$ direction, will it accelerate further in $+x$ or return to the origin? Explain using the force.
- e. Repeat for a tiny push in the $+y$ direction. What does this tell you about the point $(0, 0)$?
- f. Sketch the potential energy surface qualitatively, indicating the directions of force in each quadrant.

Problem 7: Particle Constrained to a Wire

A bead of mass $m = 0.05$ kg slides on a frictionless wire bent into the shape of a parabola $y = ax^2$ with $a = 0.5$ m⁻¹. The height of the wire above the ground is y , so the potential energy is $U = mgy$.

- Since the bead is constrained to the wire, its position is determined by x alone. Express U as a function of x only.
- Find $\frac{dU}{dx}$ using the chain rule: $\frac{dU}{dx} = \frac{dU}{dy} \cdot \frac{dy}{dx}$.
- The force along the wire (the tangential force) is $F_x = -\frac{dU}{dx}$. Find this force.
- If the bead is released from rest at $x = 2$ m, in which direction will it begin to move?
- Find the point where $\frac{dU}{dx} = 0$. What is special about this point?

Problem 8: Particle in a Rotating Frame (Centrifugal Potential)

In a rotating frame of reference, a particle of mass $m = 0.2$ kg experiences a centrifugal potential:

$$U_{\text{cent}}(x, y) = -\frac{1}{2}m\omega^2(x^2 + y^2)$$

where $\omega = 3$ rad/s is the angular velocity.

- Find $\frac{\partial U_{\text{cent}}}{\partial x}$ and $\frac{\partial U_{\text{cent}}}{\partial y}$.
- Find the centrifugal force $\mathbf{F}_{\text{cent}} = -\nabla U_{\text{cent}}$. Show that it points radially outward.
- Find the magnitude of this force at $(1, 2)$ m.
- If there is also a spring potential $U_{\text{spring}} = \frac{1}{2}kr^2$ with $k = 5$ N/m, find the total potential $U_{\text{total}} = U_{\text{spring}} + U_{\text{cent}}$.
- Find the net force $\mathbf{F}_{\text{net}} = -\nabla U_{\text{total}}$ at $(1, 2)$. In what direction does it point?
- Find the radius r where the net force in the radial direction is zero. This is where a particle could orbit without radial motion.

Problem 9: Gravitational Force from Two Masses

Two fixed masses $M = 10^{25}$ kg are located at $(a, 0)$ and $(-a, 0)$ where $a = 10^6$ m. A third mass $m = 1$ kg moves in their combined gravitational field. The potential energy of m is:

$$U(x, y) = -GMm \left(\frac{1}{\sqrt{(x-a)^2 + y^2}} + \frac{1}{\sqrt{(x+a)^2 + y^2}} \right)$$

where $G = 6.67 \times 10^{-11}$ N·m²/kg².

- Find $\frac{\partial U}{\partial x}$ and $\frac{\partial U}{\partial y}$. (You may leave the answer in terms of the distances $r_1 = \sqrt{(x-a)^2 + y^2}$ and $r_2 = \sqrt{(x+a)^2 + y^2}$.)
- Find the force components $F_x = -\frac{\partial U}{\partial x}$ and $F_y = -\frac{\partial U}{\partial y}$.
- Evaluate the force at the origin $(0, 0)$. What do you notice? Is this an equilibrium point?
- Evaluate the force at $(0, a)$. In what direction does it point?
- By symmetry, points on the y -axis have $F_x = 0$. Find F_y as a function of y (for $x = 0$).
- If a particle is placed at $(0, y)$ and released, will it fall toward the origin or away from it? Explain using the sign of F_y .

Bonus Challenge Problem

Problem 10: Path Independence and Conservative Forces

For the potential $U(x, y) = x^2 + y^2$:

- Find the force $\mathbf{F} = -\nabla U$.
- Compute the work done by this force along two different paths from $(0, 0)$ to $(1, 1)$:
 - Path A: Go from $(0, 0)$ to $(1, 0)$ along the x-axis, then to $(1, 1)$ along the vertical line.
 - Path B: Go along the straight line $y = x$ from $(0, 0)$ to $(1, 1)$.
- Show that the work is the same for both paths, and equals $U(0, 0) - U(1, 1) = -2$ J.
- Explain why this result is a consequence of the force being derived from a potential (conservative force).

Key Formulas from Notes

Concept	Formula
Partial derivative w.r.t x	$\frac{\partial f}{\partial x} = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h}$
Gradient (2D)	$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right)$
Gradient (3D)	$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)$
Direction of steepest ascent	∇f points in direction of greatest increase
Rate of steepest ascent	$ \nabla f $ gives the rate in that direction
Conservative force	$\mathbf{F} = -\nabla U$
Force components	$F_x = -\frac{\partial U}{\partial x}, F_y = -\frac{\partial U}{\partial y}, F_z = -\frac{\partial U}{\partial z}$
Chain rule	$\frac{df}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}$